



Day 7 Notes

MediaPipe Pose – Real Body Detection!

★ Today = huge milestone. Replaced my fixed red dot (Day 6) with REAL AI-detected body points, live, in the browser.

1 What is MediaPipe Pose?

A pretrained AI model (made by Google) that looks at a video frame and returns ~33 body landmark coordinates – shoulders, elbows, wrists, hips, knees, etc.

I don't train anything myself – I just feed it a video frame and it hands back the coordinates. This is called using a "pretrained model."

2 Loading MediaPipe (the RIGHT way)

```
<script type="module">
import { FilesetResolver, PoseLandmarker } from
  "https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.14"
</script>
```

type="module" + import = the modern way to load JS libraries, different from old <script src="..."> tags. This matters because modules let you pick exact named pieces (like FilesetResolver) instead of hoping a global variable exists.

3 Setting Up the Pose Model

```
const vision = await FilesetResolver.forVisionTasks(
  'https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.14/wasm'
)

poseLandmarker = await PoseLandmarker.createFromOptions(vision, {
  baseOptions: {
    modelAssetPath: '../pose_landmarker_lite.task'
  },
  runningMode: 'VIDEO'
})
```

This downloads and prepares the AI model (takes a few seconds the first time), ready to detect poses on video.

4

Detecting Points Every Frame

```
function detectLoop() {
  const result = poseLandmarker.detectForVideo(video,
performance.now())
  ctx.clearRect(0, 0, canvas.width, canvas.height)

  for (const point of result.landmarks[0]) {
    ctx.beginPath()
    ctx.arc(point.x * canvas.width, point.y * canvas.height, 5, 0,
2*Math.PI)
    ctx.fillStyle = 'lime'
    ctx.fill()
  }

  requestAnimationFrame(detectLoop)
}
```

- `detectForVideo(video, timestamp)` → runs detection on the CURRENT frame
- `point.x / point.y` → come back NORMALIZED (between 0 and 1)
- Multiply by `canvas.width / canvas.height` → gives real pixel position

Result: around 33 green dots tracking my shoulders, elbows, hips, knees etc. live, as I move!

5

Bug I Hit and Fixed

Error: Cannot read properties of undefined (reading 'FilesetResolver')

Cause: my first attempt used an old-style `<script src="...">` tag and guessed a global variable name that didn't actually exist.

Fix: switched to the proper documented method — `<script type="module">` with a real import statement. Lesson: when unsure how a library loads, check its actual docs/example instead of guessing variable names.



Big takeaway:

MediaPipe Pose = pretrained AI that finds body landmarks from video, no training needed. ES module imports are the modern, more reliable way to load JS libraries. Landmark coordinates are normalized (0-1) and need multiplying by canvas size. Next (Day 8): add Selfie Segmentation, to separate me from the background — the second AI piece needed before I can warp a clothing image onto these points.